

Desk Number _____
Student Number _____
Student Name _____

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School of Science and Engineering

FINAL EXAMINATION

Semester 2, May, 2020

CSC1001 Introduction to Computer Science: Programming Methodology

Examination Duration: 120 minutes

Reading Time: 10 minutes

This examination has 3 questions.

Exam Conditions:

This is a FORMAL Examination

This is a RESTRICTED OPEN BOOK Exam. Maximum of one (1) sheet of notes double sided are permitted

Materials Permitted In The Exam Venue:

Maximum of one (1) sheet of handwritten notes double sided are permitted. **NO OTHER MATERIALS PERMITTED**

Any calculators without the functionalities of programming and file storage are permitted.

Materials To Be Supplied To Students:

1 × 9 Page Answer Booklet

Question 1. (10 × 3% = 30%)

Pick the correct option in each of the following sub-questions. Note that only ONE option is correct.

1) Binary number 10011010.001 and hexadecimal number 2B3.8 equal to decimal numbers: A

- A. 154.125 and 691.5 ✓
 B. 154.125 and 707.5
 C. 308.25 and 691.5 ✗
 D. 308.25 and 707.5 ✗

$$100 \frac{1}{8} = 0.125$$

$$6 \times \frac{1}{16} = 0.375$$

$$3 \times 16^1 + 11 \times 16^0 + 2 \times 16^{-1} + 0.5$$

$$3 \times 16 + 11 + 0.25 + 0.5$$

$$49.75$$

2) Concerning Python language, which of the following statements is correct? D

- A. The python interpreter is only a tool to check grammar mistakes of python code. ✗
 B. Object-Oriented Programming is a unique feature of Python. 其他语言也有 ✗
 C. Python source files may use any extensions. .py / .pyc / .pyw / ... ✓
 D. Python is a cross-platform language: Python's programs written on Windows can also run on Linux and macOS. ✓

3) Concerning the following program, which of the following statement is incorrect? B

```
class Cat:
    def __init__(self, age=1):
        self.age = age

    def setAge(self, age):
        self.age = age

    def setName(self, name):
        self.__name = name

    def getName(self):
        return self.__name

cat = Cat(2)
print(cat.getName())
```

- A. The `__init__()` function can be invoked out of the definition of `class Cat`. ✓
 B. The `__init__()` function is a private method. ✗
 C. The attribute `self.__name` in `class Cat` is private. ✓
 D. The statement `print(cat.getName())` will raise an error. ✓

4) Concerning the following program, which of the following statements is incorrect? C

```

if x<2:
    print('Below 2')
elif x<20:
    print('Below 20')
elif x<10:
    print('Below 10')
else:
    print('Something else')
print('Finished')

```

- A. `print('Below 2')` will be executed when $x = 1$. ✓
- B. `print('Below 20')` will be executed when $x = 10$. ✓
- C. `print('Below 10')` will be executed when $x = 8$. ✗
- D. `print('Something else')` will be executed when $x = 20$. ✓

5) Concerning the following program, which of the following statements is incorrect? C

```

class A:
    def __init__(self, a, b):
        self.a = a
        self.b = b

    def multiply(self):
        x = a*b
        return x

    def add(self):
        x = a+b
        return x

```

- A. Class `A()` contains two newly defined data fields. ✓
- B. The scope of `self` is throughout the whole body of class `A()`. ✓
- C. Method `add()` will return the sum of two data fields `a` and `b`. ✗
- D. Variable `x` defined in `multiply()` and `add()` are two different local variables. ✓

6) Concerning the object in Python, which of the following statements is incorrect? B

- A. Every object in Python has an unique ID. ✓
- B. The type of an object cannot be determined automatically by Python interpreter, programmer must define its type. ✗
- C. Every variable is essentially a reference to an object. ✓
- D. The ID of an object will not be changed during the execution of the program. ✓

7) Concerning class inheritance in Python, which of the following statements is incorrect? C

- A. Inheritance enables you to define a general class and later extend it to define more specialized classes. ✓

- B. A subclass may inherit data fields and methods from its superclass. ✓
 C. Every instance of the superclass is also an instance of the subclass. ✗
 D. A subclass is not a subset of its superclass, usually it contains more information and methods than its superclass. ✓

8) Concerning the following program, which of the following statements is correct? D

```
class A:
    def __init__(self, a):
        self.a=a
```

```
class B:
    def __init__(self, b):
        self.b=b
```

```
m_list=[1, 2, 3]
```

```
obj_a=A(m_list)
```

```
obj_b=B(m_list)
```

```
m_list.reverse()
obj_a.a[0]=5
```

```
print(obj_b.b)
```

- A. The output of the statement `print(obj_b.b)` is [1, 2, 3].
 B. The output of the statement `print(obj_b.b)` is [5, 2, 3].
 C. The output of the statement `print(obj_b.b)` is [3, 2, 1].
 D. The output of the statement `print(obj_b.b)` is [5, 2, 1].

9) Concerning data structure and algorithm, which of the following statements is incorrect? A

- A. When evaluating an algorithm, we only need to measuring its running time. ✗
 B. Data structure concerns how to organize and access data. ✓
 C. An algorithm is a step-by-step procedure for performing some tasks in a finite amount of time. ✓
 D. When analyzing the running time of an algorithm, two commonly used approaches are experimental analysis and theoretical analysis. ✓

10) Concerning the binary search algorithm, which of the following statement is correct? A

- A. This algorithm is able to find out whether a target element exists in a given sequence of elements. ✓
 B. Binary search algorithm can be applied on an unsorted sequence. ✗
 C. The time complexity of binary search is $O(n)$. ✗
 D. Binary search is usually equally efficient compared with sequential search. ✗

Question 2. (10 × 4% = 40%)

Pick the correct option/s in each of the following sub-questions. Note that there may be MULTIPLE correct options for each sub-question. You will get 2 points when your answers are partially right.

11) Concerning the following program, which of the following statement/s is/are correct?

BD

```
class A:
    def __init__(self, i=1):
        self.i=i

    def p1(self):
        self.i+=1

class B(A):
    def __init__(self, j=1):
        super().__init__(2021)
        self.j=j

    def p1(self):
        self.i+=1

def main():
    b=B()
    b.p1()
    print(b.i)
    print(b.j)

main()
```

- A. Class B() has only one superclass. ✓
- B. In class B(), the initializer of A() is accessed by calling function `super().__init__(2021)`. ✓
- C. Class B() has only one data field. ✗
- D. The output of `b.i` and `b.j` are 2022 and 1. ✓

12) Concerning the class in Python, which of the following statement/s is/are correct?

ABD

- A. In Python, class and data type are equivalent. ✓
- B. A subclass can inherit or override the public data fields and methods of its superclass. ✓
- C. We cannot access private members both in and outside the class. ✗
- D. The scope of a local variable in a method is within this method, we cannot access it outside the method. ✓

13) Which of the following statements is/are correct? ABD

- A. Data are saved sequentially in a queue. ✓
- B. A singly linked list is also a tree. ✗

*linear structure*C. A tree is also a linked list. ~~α~~

D. Stack and queue can be implemented based on linked list. ✓

CD

14) Concerning the binary search algorithm, which of the following statement/s is/are correct? **CD**A. Binary search algorithm can be applied on an unsorted sequence. ~~α~~B. The time complexity of binary search is $O(n)$. ~~α~~C. The time complexity of binary search is $O(\log n)$. ✓

D. Binary search is usually more efficient than sequential search. ✓

~~AB~~15) Concerning linked list, which of the following statement/s is/are correct? **BC**A. For a singly linked list, each node has ~~two~~ *one* references except for tail. ~~α~~

B. We can identify the tail of a singly linked list as the node having None as its next reference. ✓

C. The time complexity of inserting a node at the tail of a singly linked list is constant. ✓

D. The time complexity of removing a node at the tail of a singly linked list is constant. ~~α~~ *$O(n)$*

BC

-4

BD

16) Concerning a binary tree, which of the following statement/s is/are correct? **ABD**A. In a binary tree, each node contains three references. ~~α~~

B. Each child node is labelled as being either a left child or a right child. ✓

C. A binary tree is proper if each node has either one or two children. ~~α~~

D. In a binary tree, data are saved hierarchically. ✓

AD

17) Concerning stack, which of the following statement/s is/are correct? **AD**

A. A stack can be accessed based on the "last in first out" principle. ✓

B. More than one element of a stack can be accessed simultaneously if necessary. ~~α~~C. The time complexity of Inserting an element into a stack is linear. ~~α~~

D. The time complexity of removing an element from a stack is constant. ✓

18) Concerning algorithm analysis, which of the following statement/s is/are correct? **AC**A. The big-Oh notation allows us to say that a function $f(n)$ is less than or equal to another function $g(n)$ up to a constant factor when n is large enough. ✓B. Function $5n^4 + 6n^3 + 2n^2 + 2n + 2n$ is $O(n^4)$. ✓

ABC

-4

- ~~C~~ C. The big-Oh notation can be used to characterize the running time of an algorithm in the asymptotic sense.
- D. When we analyze an algorithm, we are usually interested at its average performance regardless of the input size. α

19) Concerning recursion, which of the following statement/s is/are correct? **A**

- A. Calculating factorial function belongs to linear recursive algorithms. $\checkmark$
- B. A recursive definition should have only one base case. α
- C. Inside a function, only one recursive call can be made. α
- D. Quick sort is a recursive algorithm whose average time complexity is $O(n^2)$. α

20) Assume that the five numbers 1, 2, 3, 4, and 5 are pushed onto the stack, which of the following statement/s is/are correct?: **CD**

- A. If the numbers are pushed in the order of 1-2-3-4-5, then the first number popped will be 1. α
- B. In Python, these five numbers are essentially saved in a list. α array / linked list
- C. The stack can be implemented based on a dictionary. $\checkmark$
- D. The time complexity of popping out all elements in the stack is $O(n)$. $\checkmark$

Question 3. (8% + 10% + 12% = 30%)

Read the following programs and answer the corresponding questions.

21) Concerning the following program.

```
def Func(n):
    if n < 0:
        print('Invalid input.')
        return None
    elif n == 0:
        return 1
    else:
        a = (n + 1) * Func(n - 1)
        print(n)
        return a
print(Func(6))
```

先递归再 print

- 1) How many times function **Func()** will be called in total?
- 2) What are the outputs of this program?

Answer:

1) 7 3 points

2) 5 points

1
2
3
4
5
6
5040

$6 \rightarrow 5 \rightarrow 4 \rightarrow 3 \rightarrow 2 \rightarrow 1 \rightarrow 0$

$6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$

$5 \times 4 \times 3 \times 2 \times 1 = 120$

$4 \times 3 \times 2 \times 1 = 24$

$3 \times 2 \times 1 = 6$

$2 \times 1 = 2$

$1 \times 1 = 1$

$0 \times 1 = 0$

5040

22) Concerning the following program

```
def sort(L):
    N=len(L)
    while N>0:
        for i in range(N-1):
            #①
            N-=1
        return L

def main():
    L=[9, 4, 1, 8, 5, 6, 0, 2, 7]
    print(sort(L))

main()
```

Answer the following questions:

- 1) This program is to sort a given list with descending order (the output of the above program is [9, 8, 7, 6, 5, 4, 2, 1, 0]). Use bubble sort to implement the sort. Please fill the blank ① to complete the program (maybe more than one line of code are needed.)
- 2) What is the time complexity of bubble sorting algorithm?

Answer:

1) 6 points

```
if L[i] < L[i+1]:
    temp = L[i]
    L[i] = L[i+1]
    L[i+1] = temp
```

if $L[i] < L[i+1]$:
 $j = L[i]$
 $L[i] = L[i+1]$
 $L[i+1] = j$

2) $O(n^2)$ 4 points

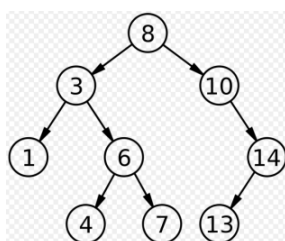
$O(n^2)$

- 23) Concerning the following program, and assume that input t will be a reference pointing to the root of a binary tree.

```
def Search(t):
    if t:
        print(t.element)
    if (t.left is None) and (t.right is None):
        return
    else:
        if t.right is not None:
            Search(t.right)
        if t.left is not None:
            Search(t.left)
```

Answer the following questions:

- 1) Which algorithm is implemented in this function? *dfs*
- 2) What is the time complexity of this algorithm? *$O(n)$*
- 3) Is this function a linear recursive or multiple recursive function? *multiple recursive*
- 4) If input t is referencing to the root of the following tree, what would be the outputs of this function?



Answer:

Depth First Search

*8
10
14
13
3
6
7
4
1*

- 1) Depth First Search 3 points
- 2) $O(n)$ 3 points
- 3) multiple recursive function 3 points
- 4) 8 3 points
- 10
- 14
- 13
- 3
- 6
- 7
- 4
- 1

END OF EXAMINATION